

GBFS Audit Catalogue

A reproducible audit of 1,509 open bike-sharing feeds across 48 countries

Raw input · transport.data.gouv.fr + MobilityData global catalogue

1,509 GBFS systems · 48 countries · ~67,000 FR stations · no semantic audit



Seven anomaly classes (A1-A7) — unified taxonomy

A1 Out-of-domain

carsharing
as BSS

46 sys.

A2 Placeholder

constant c
all stations

48 sys.

A3 Over-capacity

conditional avg
free-float

33 sys.

A4 Geospatial

transposed
>3 σ outliers

81 sys.

A5 Perimeter

overseas; rural
>50,000 km²

17 sys.

A6 Zero-capacity

c = 0 on
physical docks

14 sys.

A7 Null-capacity

c = NaN on
≥50% stations

215 sys.



Six-step purging pipeline — idempotent · reversible · logged

S1

carsharing
relabel

S2

placeholder
detect

S3

capacity
recompute

S4

geofilter
 ϕ , λ

S5

topological
 σ -clip

S6

N_{min}
threshold



Multi-source hybridisation — spatial joins · administrative aggregation

**INSEE
Filosophi**

Income, Gini

**BAAC
Cerema**

Crash density

BD TOPO

Cycle linear

**Open
StreetMap**

Facilities

**FUB
Barometer**

Perception



GBFS Audit Catalogue v1.0

DOI 10.5281/zenodo.20125460 · ODbL · 46 typed columns · Croissant + DCAT-AP

46,307

certified
stations

123

audited
systems

97

cities
covered

46

typed
columns

30.9 %

raw stations
reclassified

Headline result

Bordeaux — one A3 fix

9,921 raw entries → 225
dock-based stations

National rank

2 → 14

Syntactic standardisation is necessary but not sufficient.

An open mobility feed is a starting point for research, not a substitute for one.

Fossé & Pallares · CESI LINEACT, Montpellier · 2026